



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 15/09/2022*

PROBLEM - We observe N samples of the following discrete-time real random process:

$$X[n] = S[n] + W[n] = A \cos(2\pi F_0 n) + B \sin(2\pi F_0 n) + W[n],$$

where A and B are mutually independent Gaussian random variables with zero mean and variance σ^2 , $F_0 \in (0, 1/2)$ is a-priori known, $W[n]$ is additive white Gaussian noise (AWGN) with power σ_w^2 . Signal and noise are independent.

(1) Derive the Signal-to-Noise power Ratio (SNR) $\gamma \triangleq E\{\|\mathbf{S}\|^2\} / E\{\|\mathbf{W}\|^2\}$, where $\mathbf{S}$ and $\mathbf{W}$ are defined by $\mathbf{X} = \mathbf{S} + \mathbf{W} = [X[0] \ X[1] \ \dots \ X[N-1]]^T$. Derive the mean value function (MVF) and the autocorrelation function (ACF) of the process $X[n]$.

(2) Derive the joint probability density function (pdf) of the observed data vector $\mathbf{X}$.

(3) Apply the Orthogonality Principle to derive the optimal linear predictor of the signal of interest $S[n]$ given the observation of two samples of the process $X[n]$ at previous times, i.e. $\hat{S}[n] = h[1]X[n-1] + h[2]X[n-2]$, and express it as a function of the SNR γ . Discuss what happens to the linear predictor when $\gamma \rightarrow 0$ and $\gamma \rightarrow \infty$. Calculate numerically $h[1]$ and $h[2]$ under the assumption that $\sigma^2 = \sigma_w^2 = 1$ and $F_0 = 1/6$.

(4) Derive the Mean Square Error (MSE) $\sigma_\varepsilon^2 \triangleq E\{(S[n] - \hat{S}[n])^2\}$ of the optimal linear predictor and plot the MSE versus γ . Calculate numerically the MSE under the assumption that $\sigma^2 = \sigma_w^2 = 1$ and $F_0 = 1/6$.

Hint. $\cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos(\alpha - \beta)$



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Solution

(1) The SNR, the mean value and the ACF are:

$$\gamma \triangleq \frac{E\{\|\mathbf{S}\|^2\}}{E\{\|\mathbf{W}\|^2\}} = \frac{\sigma^2 N/2 + \sigma^2 N/2}{N\sigma_w^2} = \frac{\sigma^2}{\sigma_w^2}$$

$$\eta_x[n] = E\{X[n]\} = E\{A \cos(2\pi F_0 n) + B \sin(2\pi F_0 n) + W[n]\} = 0$$

$$R_x[m] = E\{X[n]X[n+m]\} = R_s[m] + R_w[m] = \sigma^2 \cos(2\pi F_0 m) + \sigma_w^2 \delta[m]$$

where

$$\begin{aligned} R_s[m] &= E\{S[n]S[n+m]\} \\ &= E\{(A \cos(2\pi F_0 n) + B \sin(2\pi F_0 n))(A \cos(2\pi F_0(n+m)) + B \sin(2\pi F_0(n+m)))\} \\ &= E\{A^2\} \cos(2\pi F_0 n) \cos(2\pi F_0(n+m)) + E\{B^2\} \sin(2\pi F_0 n) \sin(2\pi F_0(n+m)) \\ &= \sigma^2 (\cos(2\pi F_0 n) \cos(2\pi F_0(n+m)) + \sin(2\pi F_0 n) \sin(2\pi F_0(n+m))) \\ &= \sigma^2 \cos(2\pi F_0 m) \end{aligned}$$

(2) The pdf of $\mathbf{X}$ can be derived by observing that $X[n]$ is a Gaussian process so $\mathbf{X}$ is a Gaussian vector. In fact, $X[n] = A \cos(2\pi F_0 n) - B \sin(2\pi F_0 n) + W[n]$: since A , B and $W[n]$ are Gaussian distributed random variables, so also $X[n]$ is a Gaussian random variable. Similarly we can prove that $\mathbf{X}$ is a Gaussian random vector: $\mathbf{X} \in N(\boldsymbol{\eta}_x, \mathbf{C}_x)$, where $\boldsymbol{\eta}_x = E\{\mathbf{X}\} = \mathbf{0}$ and $\mathbf{C}_x = \mathbf{C}_s + \mathbf{C}_w = \mathbf{C}_s + \sigma_w^2 \mathbf{I}$, where

$$[\mathbf{C}_s]_{n,k} = E\{S[n-1]S[k-1]\} = R_s[n-1, k-1] = \sigma^2 \cos[2\pi F_0(n-k)], \quad n, k = 1, 2, \dots, N$$

e.g. on the main diagonal we have $[\mathbf{C}_s]_{n,n} = \sigma^2, \quad n = 1, 2, \dots, N$

(3) To find optimal linear predictor in AWGN, we have to apply the Orthogonality Principle: the estimation error $\varepsilon[n] = S[n] - \hat{S}[n]$ should be orthogonal to the data $X[n-1]$ and $X[n-2]$ used to perform the prediction

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$$\min_{\{h[n]\}} \sigma_{\varepsilon}^2 = E\{\varepsilon^2[n]\} \Rightarrow E\{\varepsilon[n]X[n-m]\} = 0, \quad m=1,2$$

$$\begin{aligned} E\{\varepsilon[n]X[n-m]\} &= E\{(S[n] - \hat{S}[n])X[n-m]\} = E\left\{\left(S[n] - \sum_{k=1}^2 h[k]X[n-k]\right)X[n-m]\right\} \\ &= E\{S[n]X[n-m]\} - \sum_{k=1}^2 h[k]E\{X[n-k]X[n-m]\} \\ &= R_{SX}[-m] - \sum_{k=1}^2 h[k]R_X[m-k] = 0, \quad m=1,2 \end{aligned}$$

Taking into account that in our case $R_{SX}[-m] = R_S[m]$, we finally came up with the linear system of equations:

$$\sum_{k=1}^2 h[k]R_X[m-k] = R_S[m], \quad m=1,2$$

that can be expressed in matrix format:

$$\begin{bmatrix} R_X[0] & R_X[1] \\ R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_S[1] \\ R_S[2] \end{bmatrix}$$

which in our case becomes:

$$\begin{bmatrix} \sigma^2 + \sigma_w^2 & \sigma^2 \cos[2\pi F_0] \\ \sigma^2 \cos[2\pi F_0] & \sigma^2 + \sigma_w^2 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} \sigma^2 \cos[2\pi F_0] \\ \sigma^2 \cos[4\pi F_0] \end{bmatrix}$$

$$\sigma^2 \begin{bmatrix} 1+1/\gamma & \cos[2\pi F_0] \\ \cos[2\pi F_0] & 1+1/\gamma \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \sigma^2 \begin{bmatrix} \cos[2\pi F_0] \\ \cos[4\pi F_0] \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 1+1/\gamma & \cos[2\pi F_0] \\ \cos[2\pi F_0] & 1+1/\gamma \end{bmatrix}^{-1} \begin{bmatrix} \cos[2\pi F_0] \\ \cos[4\pi F_0] \end{bmatrix}$$



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$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \frac{1}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]} \begin{bmatrix} 1+1/\gamma & -\cos[2\pi F_0] \\ -\cos[2\pi F_0] & 1+1/\gamma \end{bmatrix} \begin{bmatrix} \cos[2\pi F_0] \\ \cos[4\pi F_0] \end{bmatrix}$$

$$h[1] = \frac{(1+1/\gamma)\cos[2\pi F_0] - \cos[2\pi F_0]\cos[4\pi F_0]}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]}$$

$$h[2] = \frac{(1+1/\gamma)\cos[4\pi F_0] - \cos^2[2\pi F_0]}{(1+1/\gamma)^2 - \cos^2[2\pi F_0]}$$

$$\lim_{\gamma \rightarrow 0} h[1] = 0, \quad \lim_{\gamma \rightarrow 0} h[2] = 0 \quad \Rightarrow \quad \lim_{\gamma \rightarrow 0} \hat{S}[n] = 0 = E\{S[n]\}$$

which is the MMSE estimator in the absence of useful observed data, i.e. observed data correlated with the r.v. $S[n]$ we want to estimate. In this case the MSE is equal to the variance of $S[n]$, i.e. $\text{MSE} = \sigma^2$.

$$\lim_{\gamma \rightarrow \infty} h[1] = \frac{\cos[2\pi F_0] - \cos[2\pi F_0]\cos[4\pi F_0]}{1 - \cos^2[2\pi F_0]} = \frac{\cos[2\pi F_0](1 - \cos[4\pi F_0])}{\sin^2[2\pi F_0]} = 2\cos[2\pi F_0]$$

$$\lim_{\gamma \rightarrow \infty} h[2] = \frac{\cos^2[2\pi F_0] - 1}{\sin^2[2\pi F_0]} = -1 \quad \Rightarrow \quad \lim_{\gamma \rightarrow 0} \hat{S}[n] = 2\cos[2\pi F_0]X[n-1] - X[n-2]$$

which is the (pure) linear predictor for the predictable sinusoidal process, in the absence of additive noise.

$$\text{Assumption: } \sigma^2 = 1, \sigma_w^2 = 1, F_0 = 1/6 \quad \Rightarrow \quad \gamma = 1, \quad h[1] = 0.3333, \quad h[2] = -0.3333.$$

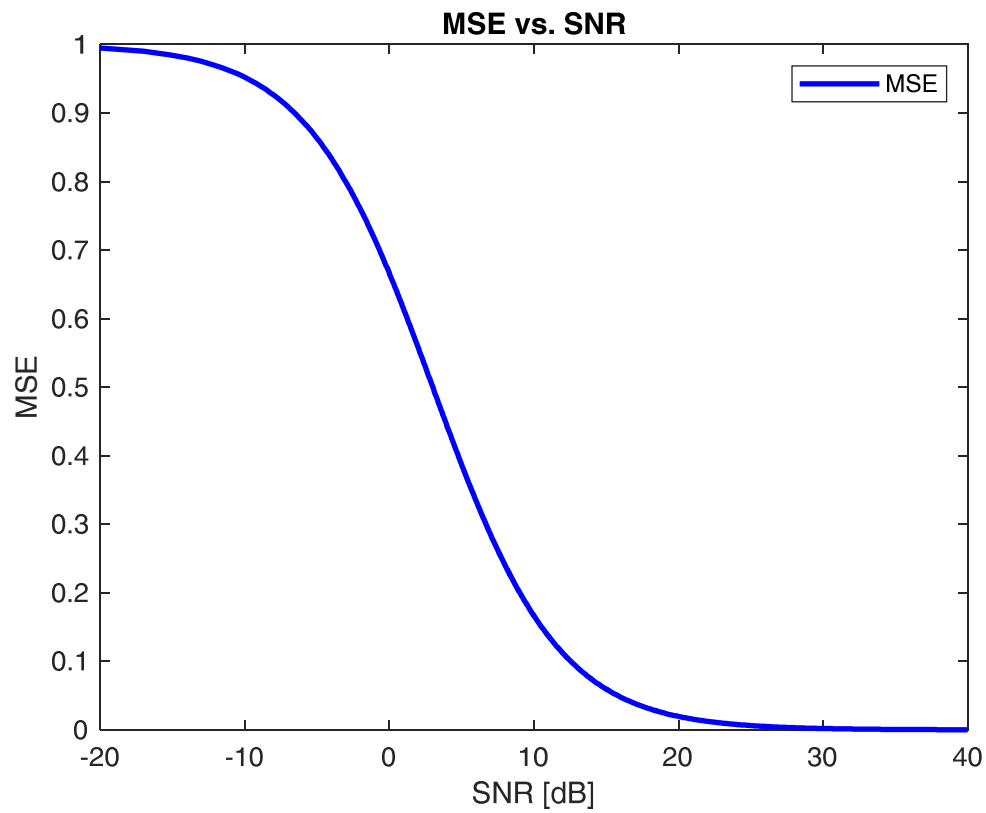
(4) The MSE is obtained by applying again the Orthogonality Principle:

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{\varepsilon[n]S[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{S^2[n]\} - \sum_{k=1}^2 h[k]E\{S[n]X[n-k]\} \\ &= R_S[0] - \sum_{k=1}^2 h[k]R_S[k] \\ &= \sigma^2 (1 - h[1]\cos[2\pi F_0] - h[2]\cos[4\pi F_0]) \end{aligned}$$



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Assumption $\sigma^2 = 1$, $\sigma_w^2 = 1$, $F_0 = 1/6 \Rightarrow \Rightarrow \gamma = 1$, $\sigma_\varepsilon^2 = 0.6667$